

# KZN Election Turnout Predictor (2000–2026)

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## 1. What the System Offers

The **KZN Election Turnout Predictor** is an interactive, executive-grade decision-support system designed to forecast ward-level voter participation for South Africa's 2026 Local Government Elections across all 921 wards in KwaZulu-Natal, with full longitudinal exploration spanning 2000 to 2026.

Live Cloud Application: [SPU Election Turnout Predictor KZN](#)

### Key Capabilities:

- **Democratic Participation & Population Coverage Indicator:** A dedicated multi-metric banner comparing **Overall Resident Population** (StatsSA baseline), **Registered Voters on Roll**, **Active Ballots Cast (Votes)**, and the **Non-Voting Population Gap**. Dynamically recalculates across statewide (**12.4M** pop), district, municipal, and ward filter levels, with transparent data governance disclosures.
  - **Analytical Justification: "Who is Not Voting, and Where Are They Located?":** Integrated analytical briefing diagnosing non-voting cohorts (disaffected youth aged 18–29, informal settlement residents facing municipal service failure, and deep rural subsistence households) and mapping their geographic hotspots (eThekweni peri-urban belt, northern traditional rural corridor, and declining industrial midland towns).
  - **Dual Temporal Selection Mode (Single Year & Year Range):** Choose between **Single Year** (instant evaluation of individual election cycles: 2026 Projected, 2021, 2016, 2011, 2006, 2000) or **Year Range** (custom temporal windows like **2011 to 2022** or **2000 to 2026** with automatic cycle detection and multi-cycle ballot accumulation).
  - **Stakeholder Review & National Scaling Feedback System:** Built-in evaluation modal dialog allowing electoral practitioners—including **IEC officials, registered voters, journalists, academic researchers, and data scientists**—to rate model performance out of 5 stars, provide qualitative feedback, and automatically dispatch evaluations directly to us by email. This feedback directly drives our architectural roadmap for expanding the predictive platform to national South African coverage (4,468 wards across all 9 provinces).
  - **Dedicated "Number of Votes" Forecasting Feature:** Converts turnout percentages into raw ballot volume forecasts (**2.7M–3.6M** votes from **5.4M–6.0M** registered voters in 2026), providing operational planners with exact ballot paper requirements.
  - **Cascading Geographic Filtering:** Seamless drill-down across all 11 District Councils, 54 Local Municipalities, and 921 Wards.
  - **Context-Aware Zero-Result Filter Fallback:** When a chosen party holds 0 wards in a selected municipality (e.g., DA in Nkandla), the system explains why by naming the actual plurality holders in that municipality (e.g., IFP and ANC) and preserves the geographical view.
  - **Authentic Continuous Socioeconomic Indicators:** Dynamically calculates Unemployment Rate, Poverty Index, and Service Delivery Rating (continuous 1-decimal scale, e.g. 4.8 to 8.2 out of 10) across active filters.
  - **Solid Party Color Cartography:** Interactive map displaying wards filled with official party colors (**ANC Green #007A3D**, **IFP Gold #D99B00**, **DA Blue #005BA6**, **MK Charcoal #222222**, **EFF Crimson #C00000**) with 6 indicator overlay modes.
  - **Multi-Chart Analytical Engine:** Instant toggling between Turnout Trends (2000–2026), Municipal Turnout Rankings (Horizontal Bar), Distribution Histograms, Socioeconomic Driver Scatter Plots, and Turnout Shift Categories.
  - **Dual Data Export Engine:** One-click CSV downloads of the filtered selection or the full statewide 921-ward 2026 master dataset.
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## 2. Technical Methodology, Tools, and Justification

| Layer                     | Technology                             | Justification                                                                                                                                                                                                                            |
|---------------------------|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Predictive Modeling       | Scikit-Learn (Random Forest Regressor) | Selected for its proven ability to model non-linear relationships between socioeconomic distress and voter turnout without artificial linearity assumptions. Successfully reduced held-out test error by 10.4% MAE over naive baselines. |
| Data Engine               | Python, Pandas, NumPy                  | Vectorized feature fusion, longitudinal panel harmonization (2000–2021), and sub-millisecond calculation of multi-cycle cascading filter aggregations.                                                                                   |
| Geospatial & Visuals      | Plotly Express & Graph Objects         | Interactive client-side map rendering, dynamic hover tooltips, and presentation-grade charts without requiring heavy GIS backends.                                                                                                       |
| User Interface            | Streamlit                              | Enterprise executive dashboard with responsive state management, clean corporate typography, and zero-clutter decision cards.                                                                                                            |
| Deployment Infrastructure | Streamlit Community Cloud              | Deployed on managed container infrastructure connected to Git for continuous integration, real-time availability, and encrypted HTTPS delivery.                                                                                          |

## 3. Data Sources and Governance

1. Electoral Commission of South Africa (IEC):
  - Ward-level results across 5 municipal election cycles (2000, 2006, 2011, 2016, 2021).
  - Certified voter registration rolls and voting district counts.
2. Statistics South Africa (StatsSA):
  - Census demographic resident populations (enumerated at municipal and district tiers).
  - Quarterly Labour Force Survey (QLFS Q1 2024) employment and youth absorption metrics.
  - Sub-place settlement classifications (urban, traditional, farm).
3. Poverty & Infrastructure Metrics:
  - Lower-Bound Poverty Line (LBPL) headcount and poverty gap assessments.
  - Municipal Infrastructure Access and continuous Service Delivery satisfaction ratings.

## 4. Operational Value for Stakeholders

- Electoral Administrators (IEC):** Translates turnout forecasts into exact ballot requirements per ward to prevent ballot shortages and queue bottlenecks.
- Municipal Planners:** Identifies wards where chronic service delivery failure correlates directly with democratic withdrawal.
- Civic Organizations:** Guides targeted youth voter registration drives and civic education campaigns to the wards at greatest risk of voter apathy.